

SESSION 01

Welcome to

THE TINKERING LAB

Build. Break. Learn. Repeat.

This Is NOT a Normal Class.

No memorising. No textbooks. No sitting quietly.



**You will BUILD
things.**



**You will BREAK
things.**



**You will LEARN
from both.**

"In this lab, mistakes are not failures — they are data."

What is STEAM?

Five subjects — one powerful way of thinking.

S

Science

Understand how nature works

T

Technology

Build tools that solve problems

E

Engineering

Design & test real solutions

A

Arts

Think creatively, design beautifully

M

Mathematics

The language of everything

These are NOT separate subjects — they work together in EVERY real-world project.

What is Tinkering?

Tinkering = Learning by DOING

- 1 Try something
- 2 See what happens
- 3 Fix what broke
- 4 Try again — better!

*"Real engineers don't get it right the first time.
They get it right eventually."*

Who tinkers?



Thomas Edison

1000 failed tries before the bulb worked



ISRO Scientists

Test rockets, fix failures, fly again



App Developers

Build → test → update → repeat

Your Lab — What's Inside?

6 tools. Infinite possibilities.



Arduino

The Brain

Programs that control hardware



Sensors

Eyes & Ears

Detect light, sound, motion, temp



Motors

Muscles

Make things spin and move



IoT

Connected Mind

Link your device to the internet



3D Printing

The Maker

Print anything you can design



Electronics

The Blood

Circuits, LEDs, resistors & more



WHAT YOU WILL BUILD THIS YEAR



Smart Light

Beginner

A light that turns on when it gets dark — automatically!



Weather Station

Intermediate

Read temperature & humidity — display it on screen!



Mini Robot

Intermediate

A robot that moves, avoids walls, follows commands.



IoT Dashboard

Advanced

Control your device from your phone, anywhere!



3D Printed Part

Intermediate

Design a custom part & physically print it.



Smart Alert System

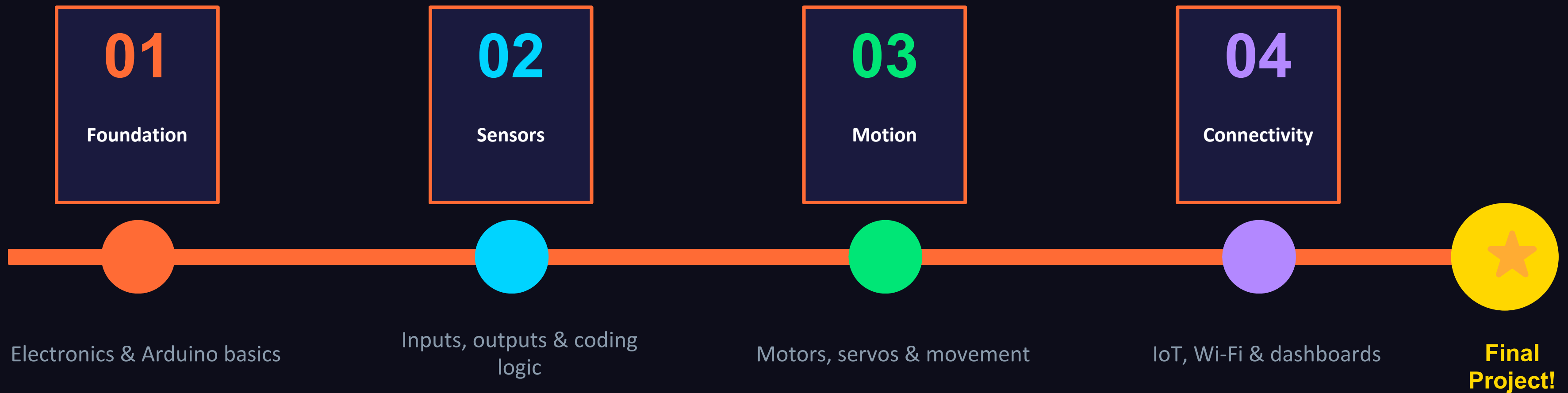
Advanced

Motion sensor that sends a notification when triggered.

These are YOUR projects. You will design, build and present them.

Your Journey — The Roadmap

Here's where you start — and where you'll go.



 Every session builds on the last. By the end, you will present a project YOU designed from scratch.

TODAY'S SESSION — WHAT HAPPENS IN THE NEXT 75 MINS



0–10 min

Introductions

Meet the lab, meet each other, set the vibe



10–25 min

STEAM + Tinkering

What are we actually doing here?



25–40 min

Explore the Lab

See the tools — Arduino, sensors, 3D printer



40–60 min

Hands-On Activity

Connect an LED circuit on a breadboard!



60–70 min

Discussion + Q&A

What broke? What worked? What did you learn?



70–75 min

Sneak Peek: Next Class

What amazing thing comes next session?

Quick Electronics Primer

Just 3 concepts — understand these and circuits make sense.

V

Voltage

Volts (V)

💧 *Water pressure*

The force that pushes electricity. Higher voltage = stronger push.

I

Current

Amps (A)

💧 *Water flow*

How much electricity flows. Think of it as the amount of water.

R

Resistance

Ohms (Ω)

💧 *A narrow pipe*

Slows down the flow. Resistors protect components from too much current.

You DON'T need to memorise these today — you will feel them when you build.

SAFETY RULES — NON-NEGOTIABLE

The lab is safe — as long as we follow these rules together.



No short circuits

Never connect + directly to – without a component in between.



Ask before connecting power

Always check with the teacher before plugging in or powering on.



Handle tools carefully

Soldering irons, cutters and blades — use them properly or ask for help.



Respect the components

They're shared by everyone. Handle with care, return to the right place.



Stop if something looks odd

Burning smell, sparks or heat? Disconnect immediately and tell the teacher.

Lab Culture — How We Roll 🤝

These aren't rules — they're superpowers.



Fail Loudly

If something doesn't work, share it.
Everyone learns from your mistake.



Team First

No lone wolves. Build together, debug together, celebrate together.



Ask Questions

There are no dumb questions. Curiosity is your biggest tool here.



Iterate Always

Version 1 is never the best. Improve, improve, improve.



Leave it Better

Clean your workspace. Put components back. Respect shared tools.



Support Each Other

If you figure something out, help the person next to you.

The best tinkerers are those who share, help, and never stop being curious.



LIVE DEMO — YOUR FIRST CIRCUIT

Turn OFF the lights. Watch the LED glow. That's where everything in technology begins.

What You Need

-  LED (any color)
-  220Ω Resistor
-  Breadboard
-  Arduino UNO or USB power
-  2 jumper wires

Steps

-  1 Place resistor on breadboard (row A)
-  2 Place LED — long leg (+) to resistor end
-  3 Connect wire: – leg to GND
-  4 Connect wire: resistor to 5V / pin
-  5 Power ON → LED glows! 🎉

🌟 *This simple LED blink is the Hello World of electronics. Every engineer starts here.*



YOUR TURN — HANDS-ON ACTIVITY

Time: 15–20 minutes | Work in pairs

Task: Build your first LED circuit on a breadboard

✓ Core Challenge

- Light up an LED
- Turn it OFF and ON by removing the wire
- Reverse the LED — what happens?



Think About It

- ? Why does the LED need a resistor?
- ? What happens if you reverse the LED?
- ? Can you control brightness? How?
- ? What if you add a second LED?



BONUS: Can you make two LEDs glow using one circuit?

Let's Talk

Before we wrap up — your thoughts matter!

1 What surprised you today?

2 What confused you? (That's great — share it!)

3 What do you MOST want to build?

4 What question are you leaving with?

Answer on a sticky note / whiteboard — or just shout it out!

👁️ SNEAK PEEK — NEXT SESSION

Session 2: Arduino — Making Things THINK

You'll write your very first **code** — and make an LED blink using a **program** instead of a wire.

Then — we add a button. And the LED obeys your command.

🧠 **YES, you will be programming hardware.**

📦 Bring to Next Class:

Curiosity ♦ A question you're still thinking about ♦ Your enthusiasm (no kit needed yet!)

See you next time — bring your questions and your energy! 🚀

You just took

your first step.

Every great engineer, inventor,
and maker started exactly where you are today.

BUILD SOMETHING AMAZING ⚡